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Spatial modelling of rainfall trends using satellite datasets and geographic information system

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ABSTRACT
This study developed a standard methodology for identifying spatial trends using satellite-based raster datasets. It involves the novelty of exploring the capabilities of a geographic information system in implementing the procedures of three trend tests, the Spearman rank order correlation (SROC) test, the Kendall rank correlation (KRC) test and the Mann-Kendall (MK) test, on raster datasets of the Tropical Rainfall Measuring Mission at 0.25° × 0.25° resolution. Comparative evaluation of the three tests revealed fair agreement of a major part of the test results for pre-, post- and non-monsoon and one-day maximum rainfall. Also, similar results from KRC and MK tests were obtained over a considerable area for annual, monsoon and monthly maximum rainfall. These findings suggest the importance of selecting the appropriate test depending on rainfall magnitudes at the chosen time scale and emphasize the robustness of the KRC and MK tests.

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1 Introduction

Rainfall is one of the key components of the hydrological cycle and is severely affected by the changing climate globally (Machiwal *et al.* 2016). Trends in rainfall series are generally determined using historical datasets, which are traditionally recorded from networks of *in situ* raingauge stations. Obtaining rainfall data from raingauge networks is quite expensive, especially in arid regions where rainfall is scant (Pombo and De Oliveira 2015). However, there are standard rainfall datasets produced by combining measurements taken from satellite-based sensors and ground observations in order to make data available for a gridded land surface on a regular basis, e.g. the Climate Research Unit (CRU 2000), the Tropical Rainfall Measuring Mission (TRMM) (Huffman *et al.* 1995, 2007), Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks (PERSIANN) (Sorooshian *et al.* 2000), the Climate Prediction Center (CPC) Morphing technique (CMORPH) (Joyce *et al.* 2004), the Global Precipitation Climatology Project (GPCP) (Schneider *et al.* 2010), the National Centers for Environmental Prediction/National Center for Atmospheric Research (NCEP/NCAR) reanalysis dataset (Kanamitsu *et al.* 2002), and the Water and Global Change (WATCH) project (Weedon *et al.* 2011). These satellite–gauge rainfall products are quite popular due to their availability at high spatial resolution (up to grid size of 1° × 1°), regular updating and closeness to the measured raingauge data (Li *et al.* 2014). The limitation of these datasets is their availability only for shorter periods, beginning in 1997.

In India, several studies have examined rainfall trends based on point measurements from rainfall stations (Guhathakurta *et al.* 2011, Sonali and Kumar 2013, Goyal 2014, Pingale *et al.* 2014, Sarkar *et al.* 2015, Machiwal and Jha 2016, Machiwal *et al.* 2016); however, trend studies dealing with satellite-based datasets are comparatively limited (e.g. Goswami *et al.* 2006, Rajeevan *et al.* 2008, Krishnamurthy *et al.* 2009, Kumar and Jain 2010, Kumar *et al.* 2010, Jain and Kumar 2012, Rathore *et al.* 2013), and they usually involved gridded rainfall datasets of 1° × 1° spatial resolution. The rainfall data at such a spatial scale may not be adequate for understanding local trends at the scale of an entire Indian state (province) or a district.

The nonparametric Mann-Kendall (MK) test has been employed for detecting rainfall trends in many studies (e.g. Yu *et al.* 2002, Liu *et al.* 2008, Qin *et al.* 2010, Tabari and Talaee 2011, Machiwal and Jha 2016, Machiwal *et al.* 2016). The Kendall rank correlation (KRC) and MK tests are equally powerful in detecting trends in time series (Machiwal and Jha 2008). Nonparametric tests are

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considered advantageous over parametric tests, as the former are free from the assumption of the presence of normality, which does not remain valid for most rainfall series (Machiwal and Jha 2016). However, one of the major assumptions of nonparametric tests is independence of the data, which is generally not observed, as most rainfall time series exhibit some sort of serial correlation or persistence (Koutsoyiannis and Montanari 2007). Some studies have reported that the MK test is not robust against the presence of serial correlation in time series (e.g. Yue *et al.* 2002), and pre-whitening of the data is required (Von Storch 1995). However, in another study, Yue and Wang (2002) extended the works of Yue *et al.* (2002) and emphasized that when a trend does exist in a time series, pre-whitening is not suitable for eliminating the influence of serial correlation on the MK test. Also, recent studies do not advocate the pre-whitening approach as it has been demonstrated clearly that pre-whitening cannot really improve trend identification when using the MK test, and may cause the wrong results to be obtained (e.g. Sang *et al.* 2014). An alternative to pre-whitening of the data is modification of the trend test to account for the effect of serial correlation (Hamed 2008).

Many past studies have used a satellite dataset for detection of trends and variability in rainfall. However, those studies determined trends for a cumulative area of certain pixels, e.g. polygons, zones, regions, by aggregating the rainfall values of the pixels falling in the considered area (Kumar and Jain 2010, Joshi and Pandey 2011, Oguntunde *et al.* 2011, Das *et al.* 2014, Diem *et al.* 2014, Manzanas *et al.* 2014, Hoscilo *et al.* 2015, Bal *et al.* 2016). In most studies, trends in the aggregate rainfall data were detected in a non-spatial manner to represent areas by treating them as points, and after analysis the results were passed to a geographic information system (GIS) for spatial presentation (Wagesho *et al.* 2013, Westra *et al.* 2013, Oza and Kishtawal 2014, Pingale *et al.* 2014, Taxak *et al.* 2014). It is further revealed from the literature that there have been relatively few studies where trends were analysed in a spatially distributed manner in the GIS platform (Gokmen *et al.* 2013, Asadieh and Krakauer 2015). No such past studies, involving spatially distributed trend analysis, applied statistical trend tests systematically; the preliminary assumption of non-persistence or absence of serial correlation of the datasets was not tested. Therefore, this study proposes, for the first time, a standard methodology for analysing the spatially distributed temporal trends in a GIS using satellite-based rainfall datasets in a comprehensive manner.

The novelty of the study involves exploring the GIS capabilities in implementation of multiple trend identification tests, i.e. KRC, Spearman rank order correlation (SROC) and MK with modified test statistics, to remove effect of serial correlation directly on the raster maps of the rainfall distribution with spatial resolution of 0.25° × 0.25°. The use of multiple statistical tests for trend detection is highly recommended for hydrological parameters (Machiwal and Jha 2008, Sonali and Kumar 2013). The proposed methodology is simple in nature and may have great practical utility in analysing trends using satellite data within a short time frame. The methodology can also be used to determine distributed trends in rainfall raster generated from spatial interpolation of point rainfall data. Moreover, the methodology is demonstrated through a case study for Gujarat State, India, where rainfall trends are detected by applying the suggested procedure.

2 Materials and methods

2.1 Study area description

The study area, Gujarat State, is situated in western India (Fig. 1) and is bounded by the Arabian Sea to the west and southwest. The climate of the area is tropical and dry, with great spatial diversity in climatic conditions. The northwest part of the study area has an extreme climate, with summers that are extremely hot, while the winters are not too cold. During winter (November–February), the climate remains mild, pleasant and dry, with mean temperatures ranging from 29°C during daytime to 12°C at night. Most parts of Gujarat experience temperatures varying from 6°C to 45°C (Ray *et al.* 2009). During summer (March–May), the area experiences an extremely hot and dry climate. Rainfall generally occurs when the southwest monsoon arrives in the area and, accordingly, the entire year can be divided into four seasons: winter (January–February), pre-monsoon (March–May), monsoon (June–September) and post-monsoon (October–December). The mean annual rainfall in the northern and southern parts of the area varies over the ranges 510–1020 mm and 760–1520 mm, respectively, whereas the southern highlands of Saurashtra and the Gulf of Cambay experience about 630 mm of rainfall, and the remaining parts of Saurashtra receive less than 630 mm of rainfall (Kumar *et al.* 2014). The northwest portion of the area is driest, receiving the

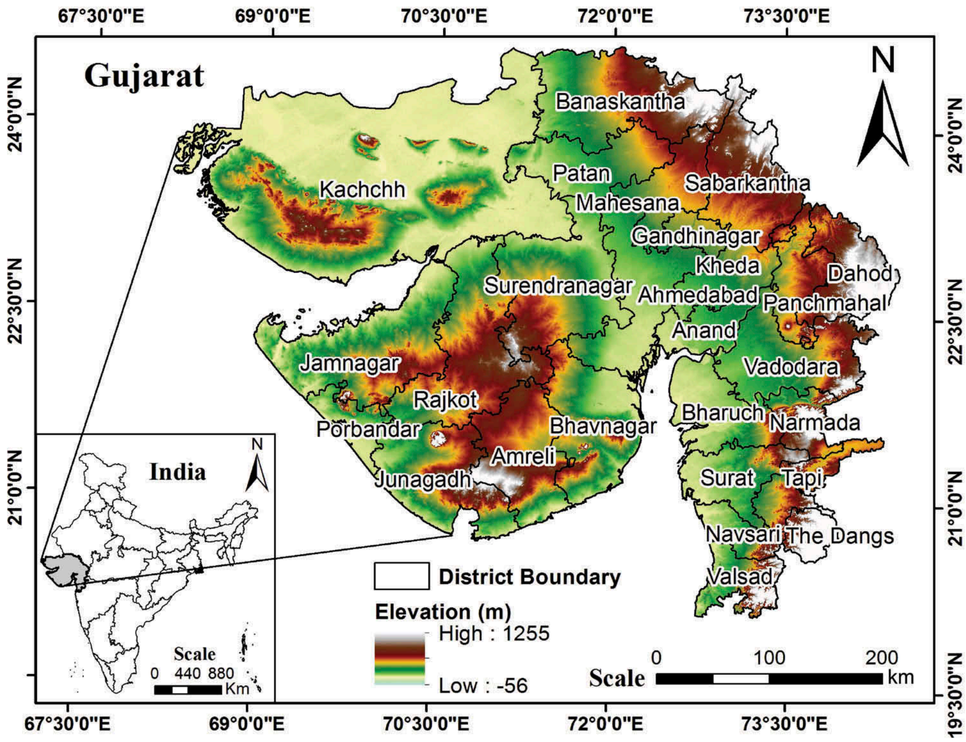

Figure 1. Location map of the study area showing the districts of Gujarat State, India.

minimum annual rainfall of 250–350 mm (Ray *et al.* 2009).

2.2 Rainfall gridded data from TRMM

The TRMM datasets consist of products generated for studying precipitation in the tropics. These products include observations of radiance, microwave temperature, radar reflectivity, rainfall rate, vertical rainfall profile, and convective and stratiform heating (Huffman *et al.* 2007). The TRMM was launched on 27 November 1997 and decommissioned on 15 April 2015. It re-entered Earth's atmosphere in June 2015 (http://disc.sci.gsfc.nasa.gov/TRMM). Because the number of days in each month differs, mm/h is used for easy comparison between the months. Both daily and hourly averages of monthly data (3B42; Version 7) are downloaded from the TRMM website (TRMM 2015). All the data (January 1998–October 2014) are downloaded in the Network Common Data Form (NetCDF) of 0.25° × 0.25° resolution. The NetCDF is a set of software libraries and self-describing, machine-independent data formats that support the creation, access and array oriented scientific data. This NetCDF data is imported as a raster layer using the Make NetCDF Raster Layer tool in GIS software. Since the 3B42 product gives the hourly average for a particular month, it has been converted into monthly rainfall by multiplying the hourly rain rate by the total hours in that month. In total, 6148 layers of daily data and 202 layers of monthly data are imported in the GIS for preparing annual, pre-monsoon (March–May), monsoon (June–September), post-monsoon (October–December), non-monsoon (excluding monsoon), one-day maximum and monthly maximum raster layers. Apart from the TRMM data, a SRTM digital elevation model (DEM) is also used to derive information about the physiography of the area. The variability in annual rainfall is assessed by computing the mean, standard deviation (SD) and coefficient of variation (CV). These statistics are developed spatially using annual rainfall (1998–2013) in raster format and district-wise means are also estimated.

2.3 Developing GIS-based methodology for identifying spatial trends using raster datasets

To the authors' knowledge, until now there has been no study explicitly describing a methodology for the analysis of spatial trends in a GIS environment. Hence, in this study, a GIS-based methodology is developed to apply statistical trend tests on raster datasets using appropriate GIS tools. Three nonparametric statistical tests, the SROC, KRC and MK tests, are selected for analysing the spatial trends in gridded rainfall data. This study utilizes the geo-processing tools by combining them in the model builder module of ArcMap software in order to optimize the number of steps to be performed in applying the tests. The developed methodology explains the proper sequence of the tools and adequate flow of different operations for detecting trends in spatial datasets using the GIS capabilities. This methodology may also help in customization of different tools to develop a single tool for analysing trends on raster datasets of high resolution. The presented methodology could easily be adopted for analysing the presence/absence of trends in other datasets over the globe. The detailed procedure for applying the tests using the developed methodology is given below.

2.3.1 Spearman rank order correlation (SROC) test

The SROC test is a nonparametric test that is usually used to check the existence of long-term trends (McGhee 1985, WMO 1988). The step-by-step procedure for applying this test in the GIS platform using raster datasets is as follows:

(1) Arrange n raster datasets (x_t) sequentially year-wise starting from the first year ($t = 1$) to the last year ($t = n$).
(2) Use the “Rank” tool to generate n “rank raster” datasets (give rank in decreasing order while running the tool in batch mode), where the first raster map contains the largest value of rainfall amount for each cell (or pixel) and the last raster map contains the smallest value for each cell.
(3) Compare the first-year raster map of the original dataset with each map of n “rank raster” datasets one-by-one by using the “Equal to” tool. If the values of two raster maps are equal for a cell, then assign 1 to that cell; otherwise assign 0. This process will generate n [1,0] raster datasets with a value of 1 for a cell in the ith raster indicating the ith rank of the rainfall amount of that cell in the first year.
(4) Use the “Iterate” tool to pick all original raster datasets one-by-one prior to repeating the last step for comparing the next “rank raster” dataset with each original raster dataset. This generates another n [1,0] raster datasets where 1 simply indicates the rank of rainfall amount that occurs in the second year for a given cell. Similarly, use the “Iterate” tool n times to compare n original raster datasets one-by-one with all n “rank raster” datasets, and generate $n \times n$ [1,0] raster datasets.
(5) Multiply the first [1,0] raster map (where 1 indicates the presence of the largest rainfall values in the first year) of the first n datasets by 1 using the “Times” tool. Similarly, multiply the second [1,0] raster map (indicating the presence of the second largest values in the first year) by 2, and then repeat this process up to the last or nth [1,0] raster map (indicating the presence of the smallest values in the first year) of the first dataset by multiplying by n. Repeat this multiplication process for the second n datasets by multiplying them from 1 to n sequentially in batch mode, and so on for all n datasets. Finally, there will be $n \times n$ raster datasets, with the first n datasets [{1,0}, {2,0}, …, {n,0}] representing only the ranks of the first year rainfall data within the spatial time series of size n, and so on.
(6) Add n raster datasets [{1,0}, {2,0}, …, {n,0}] sequentially by using the “Cell statistics” tool to generate n R_{xt} raster datasets, with $R_{xt} = 1$ for the largest x_t and $R_{xt} = n$ for the smallest x_t. Hence, the R_{xt} raster datasets basically replace the original rainfall values in successive years with their respective ranks over the period of n years.
(7) The R_{xt} raster datasets are used to compute other raster datasets d_t ($d_t = R_{xt} - t$) for $n = 1, 2, \ldots, n$ using the “Minus” tool in batch mode. The raster dataset d_t is used to calculate r_s raster datasets, as shown by (Machiwal and Jha 2012):

$$r_s = 1 - \frac{6 \sum^{t=1} n{d_t}^2}{n(n^2 - 1)} \tag{1}$$

(8) The r_s rasters are utilized to compute the test statistics given by (Machiwal and Jha 2012):

$$t_s = r_s \sqrt{\frac{n-2}{1-{r_s}^2}} \tag{2}$$

The step-by-step procedure for performing this test in a GIS platform by using raster datasets is explained by the flowchart shown in Figure 2. The significance of the presence/absence of trends is checked by comparing the computed t_s raster with the critical values at three levels of significance (l.s.): 0.01, 0.05 and 0.10. The critical values are taken from the t distribution for the chosen l.s. of α and $n - 2$ degrees of freedom. The

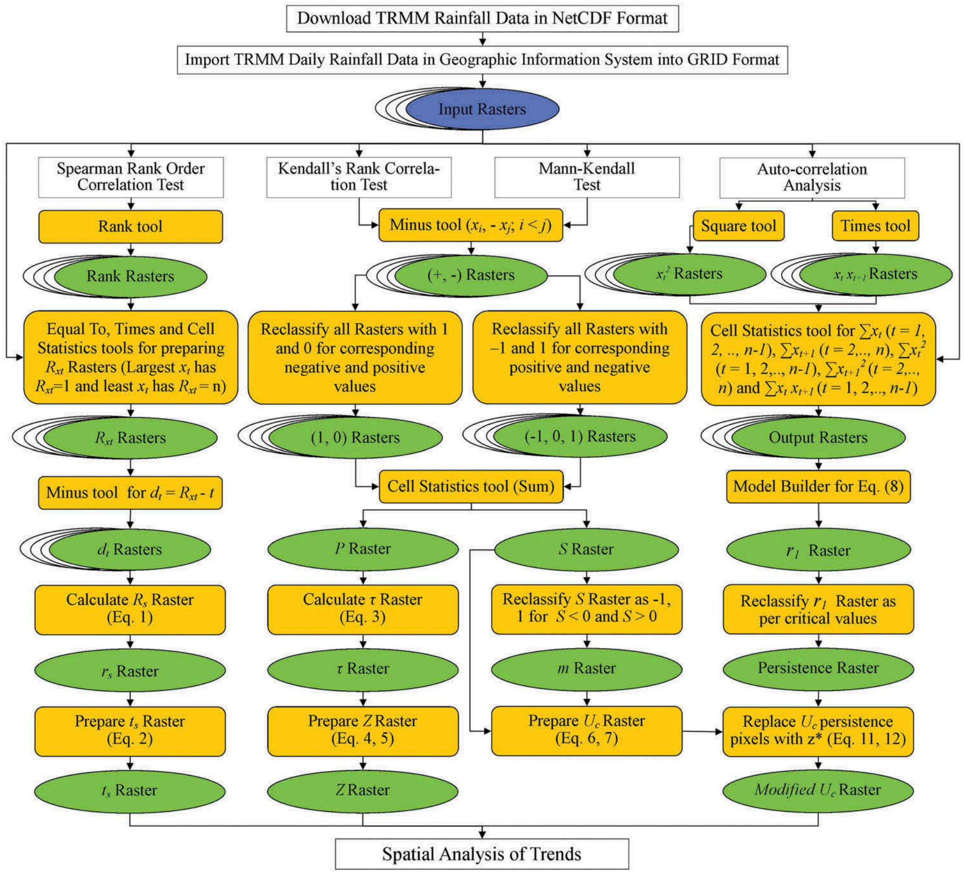

Figure 2. Flowchart depicting step-by-step procedures for applying Spearman rank order correlation, Kendall rank correlation and Mann-Kendall tests for identifying the trends in raster datasets.

critical values (n = 17) are taken as 2.602, 1.753 and 1.34 at l.s. of 0.01, 0.05 and 0.10, respectively.

2.3.2 Kendall rank correlation (KRC) test

The KRC test is one of the most preferred tests for trend detection in hydrological time series (Jayawardena and Lai 1989, Kumar 2003). The steps describing the procedure for applying this test are as follows:

(1) In sequentially arranged raster rainfall datasets x_t (t = 1, 2, ..., n) of n years, compare the first year's raster x_1 with all the subsequent years' rasters (x_2, x_3, ..., x_n) one-by-one. This may be done by taking the difference of two rasters, $x_i - x_j$ ($j > i$) in all pairs of raster datasets using the "Minus" tool and generating [+,–] raster datasets depending upon the relative values of the two rasters x_i and x_j. Repeat this process by comparing the second year's raster x_2 with all the subsequent rasters (x_3, x_4, ..., x_n) one-by-one, and so on up to comparing the x_{n-1} with the x_n raster.

(2) Use the "Reclassify" tool to classify the [+,–] raster datasets into two classes: 1 (positive values) and 0 (negative values). This step will convert [+,–] raster datasets into [1,0] raster datasets.

(3) Use the "Cell statistics" tool to add all the [1,0] rasters, and name the resultant raster dataset P, which defines the number of times the condition $x_j > x_i$ is satisfied among all pairs of raster datasets.

(4) Compute the τ raster and the variance of τ, {var (τ)}, based on the P raster dataset and the value of n using the following expressions (Kendall 1973, Machiwal and Jha 2012):

$$\tau = \frac{4P}{n(n-1)} - 1 \qquad (3)$$

and

$$\mathrm{var}(\tau) = \frac{2(2n+5)}{9n(n-1)} \tag{4}$$

(5) Finally, calculate the observed test statistics (z) as follows:

$$z = \frac{\tau}{\sqrt{\mathrm{var}(\tau)}} \tag{5}$$

The flowchart in Figure 2 shows the procedure for performing this test in a GIS platform on raster datasets. The observed values of the z statistics are compared to their critical limits of ±2.58, ±1.96 and ±1.64 at l.s. of 0.01, 0.05 and 0.10, respectively. If the observed value of the z statistic lies within these limits, the null hypothesis of no trend cannot be rejected.

2.3.3 Mann-Kendall (MK) test

The MK test is a widely used nonparametric test which explores the trend in a time series by indicating the type of trend (i.e. increasing or decreasing). The detailed procedure for implementing this test in a GIS to identify trends in raster datasets is explained as follows:

(1) Follow the first step of the KRC test to generate [+,–] raster datasets using the "Minus" tool.
(2) Classify the [+,–] raster datasets into three classes: 1 (negative values), – 1 (positive values), and 0 (zero values) using the "Reclassify" tool. This step will convert [+,–] raster datasets into [–1,0,1] raster datasets.
(3) Add all the [–1,0,1] rasters by using the "Cell statistics" tool, and name the resultant raster dataset the S raster.
(4) Identify the tied values (e_i) and their groups (g), as indicated by 0 cells in [–1,0,1] rasters. Add all the tied values in each group using the "Cell statistics" tool and generate an e_i raster for each group.
(5) Generate another m raster [1,–1] from the S raster using the "Reclassify" tool with the conditions to classify any cell value as 1 for $S < 0$, and as – 1 for $S > 0$.
(6) Combine both S and m rasters to compute the test statistic raster (U_c), defined as (Machiwal and Jha 2012):

$$U_c = \frac{S+m}{\sqrt{V(s)}} \tag{6}$$

where

$$V(s) = \frac{1}{18}\left[n(n-1)(2n+5) - \sum_{i=1}^{g} e_i(e_i-1)(2e_i+5)\right] \tag{7}$$

The observed values of the U_c statistic are compared with the critical limits of ±2.58, ±1.96 and ±1.64 at l.s. of 0.01, 0.05 and 0.10, respectively. The flowchart in Figure 2 explains the entire procedure for applying the test in a GIS framework.

2.3.4 Modified MK test for removing the effect of serial correlation

In this study, the effect of serial correlation on the MK test is removed by adopting the approach of effective sample size as proposed by Lettenmaier (1976). The presence of serial correlation is examined by performing autocorrelation analysis for all seven types of rainfall raster, i.e. annual, seasonal (pre-monsoon, monsoon, post-monsoon, non-monsoon) and extreme (one-day maximum and monthly maximum) rainfall series. The serial correlation (r_k) in the raster dataset is examined by estimating the autocorrelation coefficient/function (r_k) as follows (Machiwal and Jha 2012):

$$r_k = \frac{\sum_{t=0}^{n-k} x_t x_{t+k} - \left(\frac{1}{n-k}\right)\sum_{t=0}^{n-k} x_t \sum_{t=0}^{n-k} x_{t+k}}{\sqrt{\left[\sum_{t=0}^{n-k} {x_t}^2 - \left(\frac{1}{n-k}\right)\left(\sum_{t=0}^{n-k} x_t\right)^2\right]}\sqrt{\left[\sum_{t=0}^{n-k} {x_{t+k}}^2 - \left(\frac{1}{n-k}\right)\left(\sum_{t=0}^{n-k} x_{t+k}\right)^2\right]}} \tag{8}$$

The steps to compute r_k at lag-1 using raster datasets, shown in the flowchart (Fig. 2), are as follows:

(1) In the sequentially arranged raster rainfall datasets x_t (t = 1, 2, …, n) of n years, use the "Square" tool in batch mode over x_t (t = 1, 2, …, n) to obtain n squared rasters.
(2) Compute $x_t x_{t+1}$ (t = 1, 2, …, n – 1) rasters using the "Times" tool in batch mode; the output will be (n – 1) rasters.

(3) Use the "Cell statistics" tool to compute $\sum x_t$ (t = 1, 2, …, n – 1), $\sum x_{t+1}$ (t = 2, …, n), $\sum x_t^2$ (t = 1, 2, …, n – 1), $\sum x_{t+1}^2$ (t = 2, …, n) and $\sum x_t x_{t+1}$ (t = 1, 2, …, n – 1) in batch mode.

After computing the five input rasters defined in Step 3, use Model Builder to formulate Equation (8) and develop a new tool for completing the entire process in one click; the same tool is applied on all the other time series after defining the input parameters.

The null hypothesis (H_0) of no serial correlation is then tested by comparing the value of the lag-1 autocorrelation coefficients from the upper and lower critical limits calculated by (Machiwal and Jha 2012):

$$(r_k)_{\text{upper}} = \left[\frac{1}{(n-k)}\right]\left[-1 + z_{1-\alpha/2}\sqrt{(n-k-1)}\right] \tag{9}$$

$$(r_k)_{\text{lower}} = \left[\frac{1}{(n-k)}\right]\left[-1 - z_{1-\alpha/2}\sqrt{(n-k-1)}\right] \tag{10}$$

where $z_{1-\alpha/2}$ is the standard normal variate at l.s. of α. If $(r_k)_{\text{upper}} < r_k < (r_k)_{\text{lower}}$, the H_0 of $r_k = 0$ is rejected, indicating that the rainfall series is not purely random and some serial correlation or persistence exists; otherwise, the H_0 cannot be rejected (Machiwal and Jha 2012). The values of $z_{1-\alpha/2}$ are 1.645, 1.965 and 2.326 at l.s. of 0.10, 0.05 and 0.01, respectively (Pingale *et al.* 2014). In this study, the presence of serial correlation is examined at l.s. of 0.01 and 0.05.

In the rasters, the values of the original MK test statistics, U_c, are then replaced with the modified test statistic values for the pixels where serial correlation is found to be present. The mathematical expression proposed by Lettenmaier (1976) to calculate the modified MK test statistic value (z^*) is as follows:

$$z^* = U_c\left(\sqrt{n^*/n}\right) \tag{11}$$

where n^*/n denotes the correction factor computed for the autocorrelation function (r_1) at lag-1 from the following expression (Matalas and Langbein 1962):

$$n^*/n = \frac{1}{1 + 2\frac{r_1^{n+1} - nr_1^2 + (n-1)r_1}{n(r_1-1)^2}} \tag{12}$$

Following the above procedure, the effect of serial correlation from the MK test results is removed from all the rainfall data series.

2.4 Application of the developed methodology

The developed GIS-based methodology of the three trend detection tests is demonstrated by applying it to identify rainfall trends over Gujarat State, India, using TRMM rainfall datasets. Trends are detected in seven rainfall variables (annual, pre-monsoon, monsoon, post-monsoon, non-monsoon, one-day maximum and monthly maximum rainfall) using a spatial dataset for a period of 16 years and 10 months (January 1998–October 2014). Analysis of this study based on an approximately 17-year TRMM dataset may not be representative of long-term rainfall trends in the study area and, hence, the relative significance of the results may be enhanced by including data for additional years if they become available in the future.

2.5 Comparative evaluation of trend tests

The results of the three trend tests are spatially compared through overlay analysis in the GIS platform, in which the spatial variability of the rainfall trends under three l.s. of 0.01, 0.05, and 0.10 is explored, and areas having increasing/decreasing rainfall trends are delineated by considering similar results obtained from at least two tests. The results of the three tests are compared over space by performing union analysis based on five combinations of the similar/dissimilar results, i.e. similar results of a minimum of two tests (SROC-KRC, SROC-MK, KRC-MK and SROC-KRC-MK), and dissimilar results of all the tests.

3 Results and discussion

3.1 Spatial statistics of mean annual rainfall

The spatial statistics, mean, SD and CV, of the annual rainfall are presented in Figure 3. It is seen from Figure 3(a) that the annual rainfall is lowest (400–700 mm) in the northwestern part covered by Kachchh District, whereas it is highest (>1500 mm) in Valsad District located in the southern part of the study area. In Gujarat, 10 of the 26 districts show a mean annual rainfall of more than 1000 mm (Table 1). This indicates large spatial variations of the mean annual rainfall over the study area in comparison to previous studies reported for Gujarat (Ray *et al.* 2009, Kumar *et al.* 2010). It is further seen that the SD of the annual rainfall varies from less than 200 mm (northern part) to more than 350 mm (middle part). However, in a major part (74 644 km^2) of the study area, located in the southwest and south, the SD of the annual rainfall ranges from 300 to 350 mm (Fig. 3(b)). The lowest value of SD (229 mm) is obtained for Kachchh District

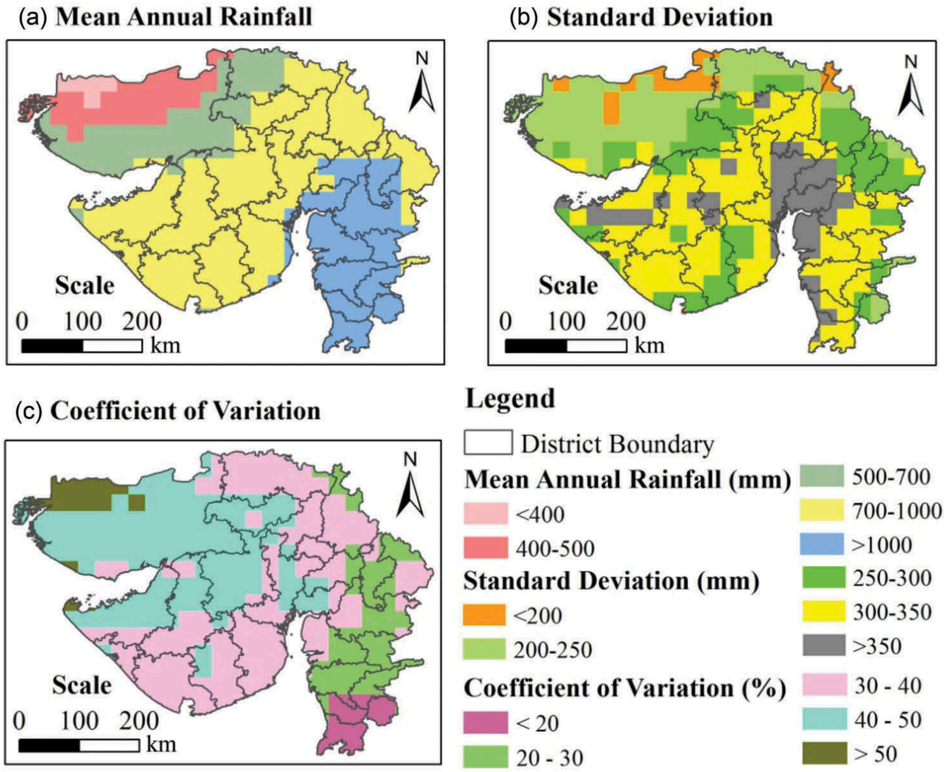

Figure 3. Spatial distribution of the (a) mean, (b) standard deviation and (c) coefficient of variation of the annual rainfall for a16-year period (1998–2013).

Table 1. Statistical summary of district-wise annual rainfall (1998–2013).

No.	District	Mean annual rainfall (mm)	Standard deviation (mm)	Coefficient of variation (%)
1	Ahmedabad	875.6	349.8	40
2	Amreli	786.1	304.8	39
3	Anand	1037.5	414.9*	40
4	Banaskantha	673.0	235.7	35
5	Bharuch	1168.6	355.7	31
6	Bhavnagar	894.1	321.1	36
7	Dahod	929.3	267.1	29
8	Gandhinagar	881.6	326.6	37
9	Jamnagar	789.1	343.8	44
10	Junagadh	843.2	312.7	37
11	Kachchh	502.8#	229.2#	46*
12	Kheda	981.1	343.7	35
13	Mahesana	832.5	320.4	38
14	Narmada	1255.3	328.9	26
15	Navsari	1742.4	332.4	19
16	Panch Mahals	1035.3	291.9	28
17	Patan	727.0	311.4	43
18	Porbandar	851.3	320.0	38
19	Rajkot	802.2	318.0	40
20	Sabarkantha	816.1	259.6	32
21	Surat	1343.8	334.6	25
22	Surendranagar	799.6	330.0	41
23	Tapi	1245.7	292.3	23
24	The Dangs	1473.9	251.3	17#
25	Vadodara	1078.0	323.6	30
26	Valsad	1882.2*	338.3	18

*Maximum; #Minimum.

and the highest (415 mm) for Anand District. The variability of the CV of the annual rainfall ranges from less than 20% to more than 50% (Fig. 3(c)). The CV of the annual rainfall is very high (> 40%) for seven districts, i.e. Rajkot, Ahmedabad, Anand, Surendranagar, Patan, Jamnagar and Kachchh (Table 1). The CV values are above 50% in northern Kachchh, and these are in close agreement with Machiwal *et al.* (2016). However, three districts situated at the southern end of the study area (Fig. 3(c)) show CV values of less than 20%, with the lowest value (17%) for The Dangs. Kumar *et al.* (2010) reported an average CV of annual rainfall for Gujarat of 31%. It is clearly seen from the above discussion that the SD and CV are highest in the northern and northwestern parts of the area, indicating large spatial variations where the mean annual rainfall is very low.

3.2 Presence of serial correlation

The serial correlation in all seven rainfall series was examined at lag-1, and the results are presented in Figure 4. It is revealed that serial correlation is present

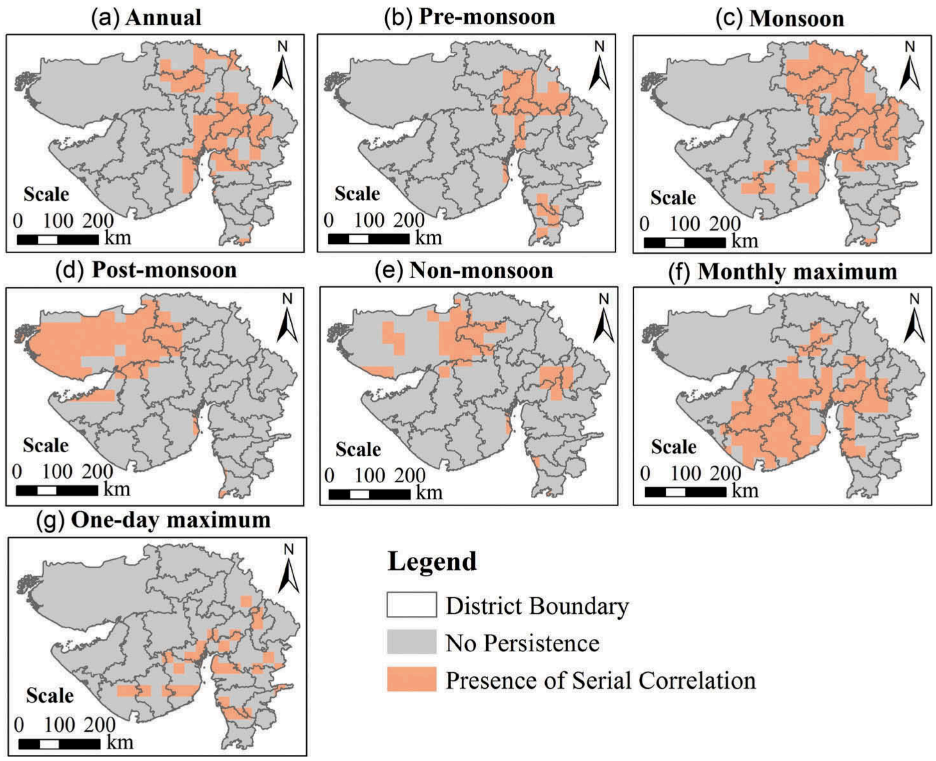

Figure 4. Distribution of serial correlation at lag-1 for different time series.

over large areas in all the rainfall series, ranging from 18 878.3 km^2 (10.2%) for one-day maximum rainfall to 60 961.1 km^2 (32.8%) in the southwest of the study area for monthly maximum rainfall (Fig. 4(f,g)). In the case of annual rainfall, the serial correlation is found mostly in the middle-northern districts, i.e. Patan, Banaskantha, Sabarkantha, Gandhinagar, Ahmedabad, Bhavnagar, Bharuch, Anand, Kheda, Panchmahal, Dahod and Vadodara (Fig. 4(a)). For pre-monsoon and monsoon rainfall, serial correlation was found to be significant over 20 525 km^2 (11%) and 57 354.1 km^2 (30.8%) of the area, respectively (Fig. 4(b,c)); for post-monsoon rainfall, serial correlation was predominant for 46 251.6 km^2 (24.9%), over Kachchh and adjoining parts (Fig. 4(d)); and for non-monsoon rainfall, serial correlation was observed for 25 703.7 km^2 (13.8%), in eastern Kachchh, western Patan and parts of Surendranagar, Rajkot, Kheda, Panchmahal, Vadodara and Navsari districts (Fig. 4(e)).

3.3 Spatial distribution of rainfall trends

3.3.1 Annual rainfall

The spatially distributed trends of annual rainfall identified by the three nonparametric tests are illustrated in Figure 5. The results of the SROC test (Fig. 5(a)) show that annual rainfall has a significant trend at 0.10 l.s. over 146 101.7 km^2 (78.6% of the study area), except for the southern portion and some pockets in the middle, southwest and northwest (Fig. 5(a)). Similarly, the rainfall trends are found to be statistically significant at 0.05 l.s. over 85 145.5 km^2 (45.8%), covering 10 districts (Kachchh, Banaskantha, Sabarkantha, Jamnagar, Rajkot, Surendranagar, Amreli, Patan, Panchmahal and Dahod). The highly significant trends at 0.01 l.s are distributed over pockets of five districts (Kachchh, Banaskantha, Sabarkantha, Jamnagar and Rajkot) covering 5745.6 km^2 (3.1%). Of the total of 26 districts, only eight were found to have non-significant rainfall trends (Table 2).

The results of the KRC test (Fig. 5(b)) indicate that annual rainfall has an increasing trend over 173 745.9 km^2 (93.5%) and a decreasing trend over 12 165.5 km^2 (6.5%). However, the increasing trends are found to be statistically significant over 84 544.1 km^2 (45.5%), 32 857.3 km^2 (17.7%) and 309.1 km^2 (0.2%) at l.s. of 0.10, 0.05 and 0.01, respectively. The significantly increasing trends at 0.05 l.s. are visible in the northwest and southwest parts of the study area (Saurashtra-Kachchh region) and in the northeast (Fig. 5(b)) comprising six districts (Kachchh, Amreli, Banaskantha, Sabarkantha, Jamnagar and Panchmahal). Similarly, significantly increasing rainfall trends have been reported by Machiwal *et al.* (2016) for the northern part of Kachchh District, Gujarat. Furthermore, decreasing trends in rainfall are seen only in the southern part of

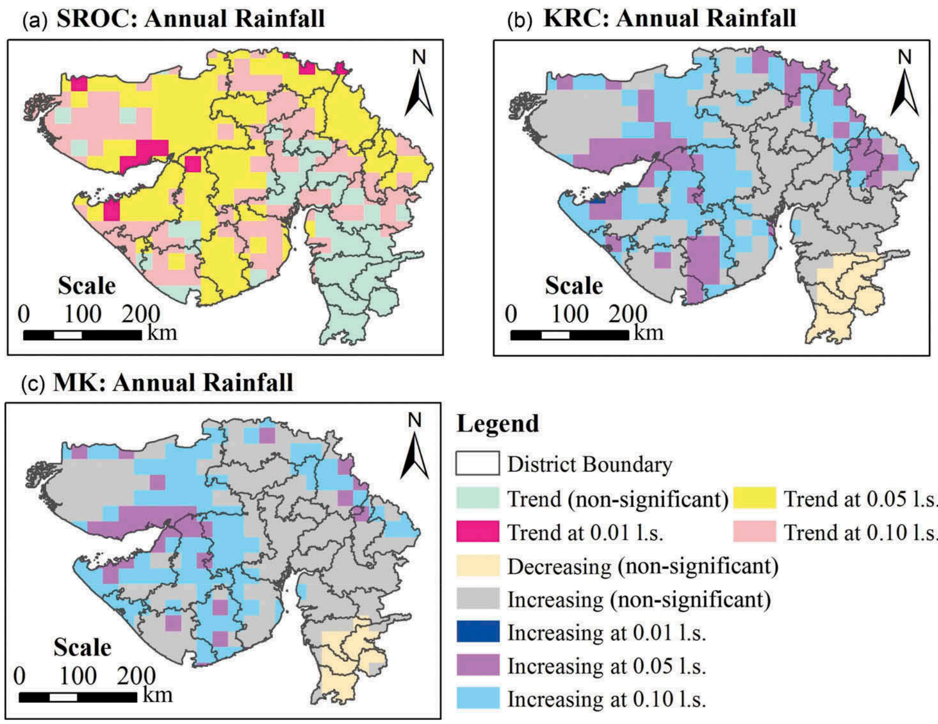

Figure 5. Spatial distribution of the annual rainfall trends identified at three levels of significance (l.s.) by (a) SROC, (b) KRC and (c) MK tests.

Table 2. Spearman rank order correlation test statistics for 26 districts of Gujarat.

District	Annual	Pre-monsoon	Monsoon	Post-monsoon	Non-monsoon	Daily maximum	Monthly maximum
Ahmedabad	1.30	0.46	1.38*	1.11	1.51*	0.51	1.93#
Amreli	1.97#	0.37	1.97#	0.51	1.18	1.37*	2.09#
Anand	1.36*	0.77	1.25	0.77	1.95#	0.85	2.13#
Banaskantha	2.02#	0.23	2.31#	0.94	1.01	0.73	1.78#
Bharuch	1.14	1.38*	0.99	0.25	1.41*	0.84	1.22
Bhavnagar	1.68*	1.05	1.48*	0.35	0.95	1.13	2.15#
Dahod	1.85#	0.52	1.79#	0.31	0.99	0.94	1.98#
Gandhinagar	1.40*	0.86	1.23	0.87	1.42*	0.59	1.43*
Jamnagar	2.03#	1.00	1.75*	0.80	1.24	1.13	1.57*
Junagadh	1.46*	0.80	1.68*	0.83	1.28	1.08	1.30
Kachchh	1.93#	0.59	2.30#	0.60	0.66	0.87	1.39*
Kheda	1.50*	0.70	1.35*	0.82	1.49*	0.63	1.32
Mahesana	1.89#	0.90	2.18	0.98	1.49*	1.08	3.23*#
Narmada	1.24	1.26	1.06	0.33	1.09	0.67	0.93
Navsari	0.27	3.50*#	0.54	0.12	1.08	0.94	0.34
Panchmahal	1.96#	0.96	1.65*	0.54	1.62*	1.13	1.87#
Patan	1.68*	0.49	2.03#	0.97	1.58*	0.89	1.95#
Porbandar	1.80#	2.02#	2.04#	0.86	1.58*	1.14	1.40*
Rajkot	1.93#	0.56	1.92#	0.82	1.49*	1.21	2.00#
Sabarkantha	2.15#	0.97	2.28#	1.09	1.30	1.28	2.55#
Surat	0.49	1.81#	0.59	0.24	1.10	1.00	0.59
Surendranagar	1.83#	0.40	1.64*	1.02	1.66*	0.66	2.31#
Tapi	0.33	1.22	0.28	0.53	1.33	1.40*	0.90
The Dangs	0.08	1.94#	0.28	0.43	1.25	1.98#	0.39
Vadodara	1.37*	1.08	1.15	0.48	1.53*	0.75	1.34
Valsad	0.29	2.81*#	0.42	0.20	0.73	0.51	0.37

* 0.10 level of significance (l.s.); # 0.05 l.s.; *# 0.01 l.s.

the study area, covering the whole of Valsad and The Dangs districts, and parts of Navsari, Tapi, Surat and Narmada districts. However, none of the decreasing trends are found significant at any level of significance. It may be seen from Table 3 that the results of the KRC test reveal a significant increase in the annual rainfall for 10 districts at 0.10 l.s.

It can be seen from Figure 5(c) that increasing trends using the MK test are present over 176 086.6 km^2 (94.7% of the study area), except in

Table 3. Kendall's rank correlation test statistics for 26 districts of Gujarat.

District	Annual	Pre-monsoon	Monsoon	Post-monsoon	Non-monsoon	Daily maximum	Monthly maximum
Ahmedabad	1.07	−0.43	1.17	−1.09	−1.65*	0.18	1.56
Amreli	1.88*	−0.24	1.81*	−0.87	−1.46	1.11	1.74*
Anand	1.15	−0.71	1.22	−0.81	−1.82*	0.71	1.76*
Banaskantha	1.72*	0.03	1.87*	−0.85	−1.16	0.60	1.48
Bharuch	0.96	−1.40	0.87	−0.36	−1.21	0.69	1.04
Bhavnagar	1.60	0.87	1.39	−0.05	−0.78	0.54	1.68*
Dahod	1.82*	−0.56	1.70*	−0.17	−1.08	0.85	1.86*
Gandhinagar	1.26	−0.82	1.19	−0.95	−1.80*	−0.49	1.07
Jamnagar	1.84*	−1.10	1.64*	−0.58	−1.18	0.96	1.42
Junagadh	1.61	−0.73	1.66*	−1.20	−1.53	0.88	1.35
Kachchh	1.69*	0.32	1.97#	−0.15	−0.40	0.71	1.16
Kheda	1.44	−0.60	1.38	−0.90	−1.49	0.43	1.07
Mahesana	1.60	−0.94	1.78*	−0.81	−1.84*	0.72	2.46#
Narmada	1.06	−1.30	0.82	−0.23	−0.97	0.16	0.89
Navsari	−0.59	−2.80*#	−0.82	0.05	−0.99	−0.74	−0.33
Panchmahal	1.89*	−0.96	1.68*	−0.63	−1.47	1.07	1.81*
Patan	1.41	−0.22	1.59	−0.80	−1.67*	0.66	1.53
Porbandar	1.94*	−2.06#	1.98#	−1.26	−1.71*	1.15	1.44
Rajkot	1.83*	−0.50	1.77*	−0.84	−1.48	1.09	1.75*
Sabarkantha	1.92*	−0.91	1.93*	−0.93	−1.59	1.12	2.00#
Surat	−0.09	−1.57	−0.24	0.03	−1.01	−0.90	0.04
Surendranagar	1.68*	−0.36	1.48	−0.89	−1.72*	0.46	1.95*
Tapi	−0.27	−1.15	−0.74	−0.41	−1.22	−1.32	−0.99
The Dangs	−0.09	−1.81*	−0.16	−0.33	−1.14	−1.84*	0.14
Vadodara	1.20	−1.09	1.04	−0.42	−1.33	0.76	1.19
Valsad	−0.41	−2.36#	−0.69	0.18	−0.70	−0.35	0.10

* 0.10 level of significance (l.s.); # 0.05 l.s.; *# 0.01 l.s.
Note: Negative and positive values denote decreasing and increasing trends, respectively.

the southern part where there are decreasing trends over 9824.8 km^2 (5.3%). Kumar *et al.* (2010) reported increasing trends in June and August and decreasing trends in July and September. The increasing trends are statistically significant at 0.10 l.s. over 74 124.6 km^2 (39.9%), mostly in the Saurashtra-Kachchh region, and in a fringe in the northeast of the study area. These findings match well with those reported by Kumar *et al.* (2010) for the Saurashtra-Kachchh region and by Machiwal *et al.* (2016) for Kachchh District. Highly significant increasing rainfall trends at 0.05 l.s. are found over 18 200.1 km^2 (9.8%). The MK test could not detect significant trends at 0.01 l.s. Significantly increasing rainfall is detected in six districts of the area at l.s. of 0.10 (Table 4).

It is clear from the above discussion that, in general, the results of the three tests agree with each other. There is close agreement between the results of the KRC and MK tests. However, the SROC test does not contain information about the type of trends (increasing or decreasing), which is clearly indicated by the KRC and MK tests. Pingale *et al.* (2015) reported similarity in the results of annual rainfall trends detected by the MK test, the MK test with pre-whitening and the modified MK test. Therefore, in this study, the results of the KRC and MK tests are preferred over those of the SROC test.

3.3.2 Seasonal rainfall

Spatial trends of the seasonal rainfall for four seasons (pre-monsoon, monsoon, post-monsoon and non-monsoon) are depicted in Figure 6. It is seen from the results of the SROC test that the trends in the pre-monsoon rainfall are statistically significant only over 36 464.5 km^2 (19.6% of the area), which mainly covers the southern part of the study area (Fig. 6(a)). The rainfall trends are highly significant in six districts, including Navsari and Valsad at 0.01 l.s. (Table 2). The KRC and MK tests show decreasing rainfall trends over 74.4% and 75% of the area, respectively, in the pre-monsoon season, with statistically significant trends over 16 009 km^2 (8.6%) and 14 455.4 km^2 (7.8%), respectively, at 0.10 l.s. In contrast, increasing trends are distributed over 47 560.6 km^2 (25.6%) and 46 473.2 km^2 (25%) for the KRC and MK tests, respectively, which are almost non-significant in the entire area. Significantly increasing trends are detected in some pockets of three districts (Kachchh, Bhavnagar and Bharuch) by both the KRC and MK tests (Fig. 6(b, c)). In contrast, statistically significant decreasing trends are found in four districts (Valsad at 0.05 l.s., Porbandar at 0.05 l.s., The Dangs at 0.10 l.s. and Navasari at 0.01 l.s.) by means of the KRC test (Table 3). The MK test also shows statistically significant decreasing trends in Navsari, The Dangs and Valsad districts at 0.05 l.s. (Table 4). It is clear from the above discussion that the results of the KRC and

Table 4. Mann-Kendall test statistics for 26 districts of Gujarat.

District	Annual	Pre-monsoon	Monsoon	Post-monsoon	Non-monsoon	One-day maximum	Monthly maximum
Ahmedabad	0.90	−0.40	0.84	−1.19	−1.72*	−0.12	1.12
Amreli	1.87*	−0.30	1.90*	−1.03	−1.56	1.28	0.91
Anand	0.53	−0.58	0.60	−1.13	−1.66*	0.63	1.09
Banaskantha	1.46	0.04	1.33	−0.85	−1.11	0.81	1.52
Bharuch	0.76	−0.68	0.77	−0.27	−0.92	0.45	0.94
Bhavnagar	1.38	0.37	1.17	−0.27	−1.14	0.37	0.97
Dahod	1.60	−0.42	1.48	−0.11	−1.15	0.93	1.96#
Gandhinagar	1.15	−0.44	0.89	−1.31	−1.85*	−0.06	1.73*
Jamnagar	1.91*	−0.39	1.72*	−0.09	−0.70	0.98	1.28
Junagadh	1.48	−0.96	1.57	−1.16	−1.42	0.45	1.03
Kachchh	1.61	0.27	1.97#	−0.14	−0.34	0.75	1.23
Kheda	0.97	−0.47	0.95	−1.07	−1.33	0.49	1.18
Mahesana	1.26	−0.67	1.11	−0.89	−1.79*	0.46	1.80*
Narmada	0.29	−1.01	−0.04	−0.29	−1.31	−0.70	−0.12
Navsari	−0.20	−2.08#	−0.48	0.02	−1.01	−0.30	−0.05
Panch Mahals	1.64*	−0.76	1.14	−0.71	−1.34	0.69	1.71*
Patan	1.12	0.12	1.12	−0.80	−1.22	0.73	1.39
Porbandar	1.79*	−1.22	1.91*	−0.83	−1.37	0.97	0.95
Rajkot	1.77*	−0.77	1.71*	−0.85	−1.48	1.10	1.16
Sabarkantha	1.67*	−0.92	1.48	−0.97	−1.73*	1.24	2.27#
Surat	−0.26	−1.47	−0.60	0.32	−0.89	−1.18	−0.21
Surendranagar	1.60	−0.42	1.49	−0.88	−1.61	0.65	1.48
Tapi	0.05	−1.47	−0.19	−0.37	−0.96	−0.99	−0.47
The Dangs	−0.14	−2.10#	−0.40	−0.26	−1.22	−1.44	0.18
Vadodara	1.12	−1.20	1.08	−0.51	−1.13	0.79	1.33
Valsad	−0.36	−2.12#	−0.69	0.17	−0.61	−0.47	0.12

* 0.10 level of significance (l.s.); # 0.05 l.s.

Note: Negative and positive values denote decreasing and increasing trends, respectively.

MK tests are very similar to each other for pre-monsoon rainfall.

In the monsoon season, the SROC test identifies statistically significant rainfall trends over 14 0074.1 km^2 (75.3% of the area) at 0.10 l.s., extending across the area except for the southern portion (Fig. 6(d)). The KRC test indicates significant trends over 39 452.8 km^2 (21.2%) and 95 463.4 km^2 (51.3%) at l.s. of 0.05 and 0.10, respectively (Fig. 6(e)), whereas the MK test shows significant trends over 27 481.4 km^2 (14.8%) and 71 379.6 km^2 (38.4%) at the same l.s., respectively. The highly significant increasing trends at 0.05 and 0.01 l.s. are concentrated over the northwest region of Kachchh, pockets of Saurashtra and northeast parts of the study area. It can be seen from Tables 3 and 4 that significantly increasing trends in monsoon rainfall at 0.10 l.s. are found in 11 and five districts by the KRC and MK tests, respectively, whereas the SROC test shows significant trends in 15 districts (Table 2) at the same l.s. The decreasing trends of monsoon rainfall are located in the southern part of the area, and are not statistically significant at any l.s. The monsoon trends are very similar to annual trends, except for some parts of western Kachchh and Patan districts, which is consistent with the findings of Pingale *et al.* (2014).

In the post-monsoon season, significant rainfall trends are not found over a large area (92.9%) by the SROC test (Fig. 6(g)). Very similar results obtained by the KRC and MK tests further illustrate that decreasing trends prevail over large areas: 15 2974.8 km^2 (82.3%) and 15 3682.5 km^2 (82.7%), respectively, in comparison to the area under the increasing rainfall trends: 32 936.6 km^2 (17.7%) and 32 228.9 km^2 (17.3%), respectively (Fig. 6(h,i)). A significantly smaller area of 68 km^2 (0.04%) is found to have statistically significant increasing rainfall trends at 0.10 l.s. by the KRC and MK tests; however, significantly decreasing trends at 0.10 l.s. are seen over 708.6 km^2 (0.38%). At district level, none of the tests reveal statistically significant rainfall trends (Tables 2–4).

The results of the three trend tests for the non-monsoon season rainfall are roughly similar to each other (Fig. 6(j,k,l)). The SROC test reveals significant trends over 73 468.4 km^2 (39.5%) at 0.10 l.s., mostly extending across the middle and southwest parts of the study area (Fig. 6(j)), whereas the KRC and MK tests give decreasing rainfall trends over 173 559.9 km^2 (93.4%), compared to 12 351.5 km^2 (6.6%) with increasing trends. Of the total areas showing decreasing rainfall trends, the results of the KRC and MK tests show that 41 357.1 km^2 (22.2%) and 42 763.7 km^2 (23%), respectively, have statistically significant rainfall

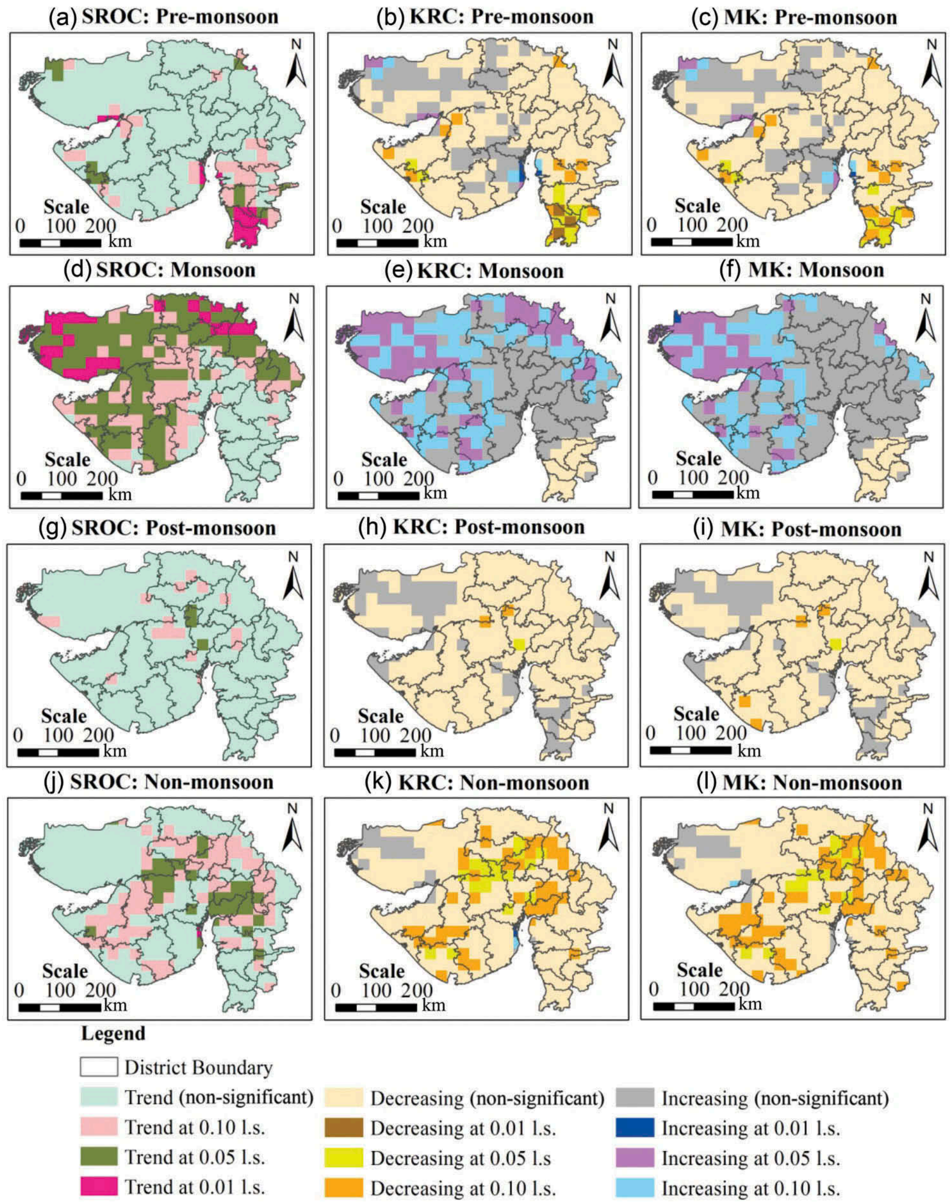

Figure 6. Spatial distribution of the rainfall trends for pre-monsoon, monsoon, post-monsoon and non-monsoon seasons identified at three levels of significance (l.s.) by SROC, KRC and MK tests.

trends at 0.10 l.s. The KRC and MK tests indicate, respectively, only seven and five of the 26 districts with a significant decrease in non-monsoon rainfall at 0.10 l.s.; in the other districts, non-significantly decreasing rainfall is observed (Tables 3 and 4).

3.3.3 Monthly and one-day maximum rainfall

The results of the three trend tests depicting the spatial trends of the monthly maximum and one-day maximum rainfall are presented in Figure 7. It is seen from Figure 7(a) that significant trends in monthly maximum rainfall are present over 125 024.3 km^2 (67.2% of the area), based on the SROC test. However, the results of both the KRC and MK tests (Fig. 7(b,c)) show increasing trends over 177 141.6 km^2 (95.3%). Decreasing trends are found in a relatively smaller area, 8769.8 km^2 (4.7%) based on both the KRC and MK tests. Significantly increasing trends at 0.10 l.s. obtained from the KRC and MK tests are found over 77 672.4 km^2 (41.8%) and 43 640.9 km^2 (23.5%), respectively, predominantly in the middle and north-eastern parts of the study area. The district-level analysis indicates that more than 60% of the area (including 16 districts) shows a significant trend of monthly maximum rainfall from the SROC test (Fig. 7(a)), and of that, highly significant trends are observed in 11 districts (Ahmedabad, Amreli, Anand, Banaskantha, Bhavnagar, Dahod, Panchmahal, Patan,

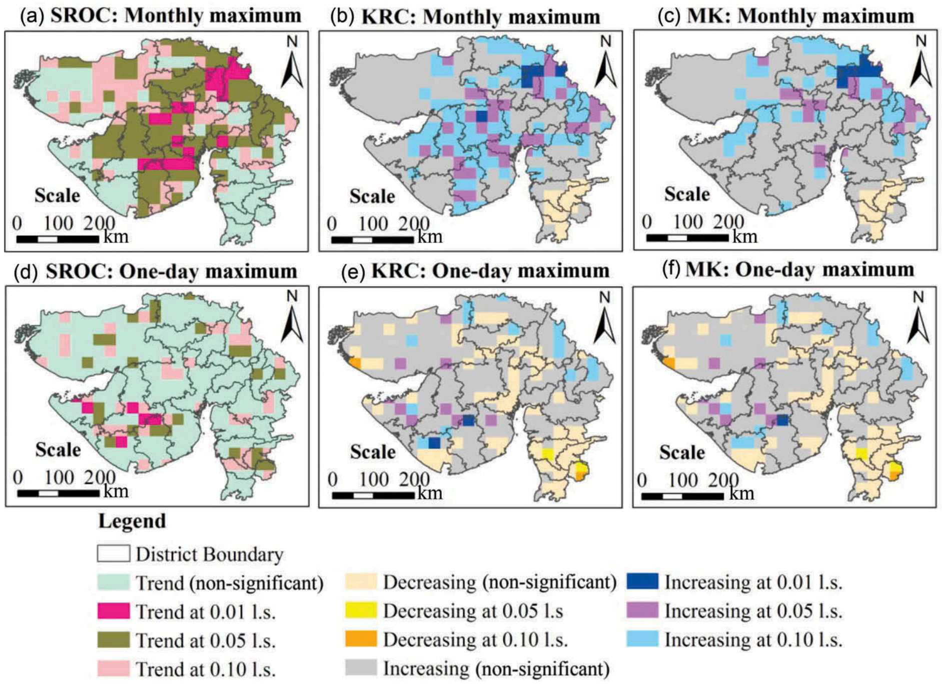

Figure 7. Spatial distribution of the rainfall trends for monthly maximum and one-day maximum rainfalls identified at three levels of significance (l.s.) by SROC, KRC and MK tests.

Rajkot, Sabarkantha and Surendranagar) at 0.05 l.s. (Table 2), as well as in Mahesana District at 0.01 l.s. The KRC test reveals significantly increasing trends at 0.05 l.s. for Mahesana and Sabarkantha districts, whereas the MK test finds a significant increase in monthly maximum rainfall at 0.05 l.s. in two districts: Dahod and Sabarkantha (Tables 3 and 4). The results of the KRC and MK tests closely match each other, and deviate slightly from the results of the SROC test.

In the case of the one-day maximum rainfall, it is observed from the results of the SROC test (Fig. 7(d)) that significant rainfall trends are distributed over small areas extending over 45 744.1 km^2 (24.6%). The results of both the KRC and MK tests show increasing trends over relatively large areas (139 036 km^2, 74.8%) compared to the area under decreasing trends (46 875.4 km^2, 25.2%). However, the increasing trends are found statistically significant at 0.10 l.s. only over 84.5 km^2 (9.1%) and 14 744.5 km^2 (7.9%), based on the KRC and MK tests, respectively. In contrast, all the decreasing trends are found non-significant at all levels of significance, excluding a very small area of 2441 km^2 (1.3%), which shows a statistically significant decreasing rainfall trend at 0.10 l.s. At district level, significant trends in one-day maximum rainfall are obtained from the SROC test in three districts: Amreli, Tapi and The Dangs (Table 2), whereas the KRC test shows significantly decreasing trends in The Dangs district only at 0.10 l.s. (Table 3) and the MK test does not find any significant trend. The assessment of trends of extreme annual daily rainfall events is crucial for adaptation planning in arid regions of India (Pingale *et al.* 2014).

3.4 Comparisons of spatially distributed trends

The results of the three trend tests were spatially compared through union analysis for five possible combinations of similar and dissimilar trends, and the results for all seven types of rainfall raster dataset are presented in Figure 8. It is seen from Figure 8(a) that, in the case of annual rainfall, the results of the KRC and MK tests are in good agreement over 104 690.6 km^2 (56.3% of the area), whereas the results of all the tests are found to be in good coherence for 57 840.8 km^2 (31.1%). In contrast, the results of all the tests show strong harmony with each over most of the area when applied to pre-monsoon (156 897.2 km^2, 84.4%) and post-monsoon (171 350.6 km^2, 92.2%) seasons (Fig. 8(b, d)). In both types of seasonal rainfall, the second dominating combination is for similar results obtained from the KRC and MK tests. Similarly, for the monsoon season, the results of the KRC and MK tests reveal spatial consistency with each other over 91 887.5 km^2 (49.4%), and the results of all the tests are close to each other for 69 581.5 km^2 (37.4%) (Fig. 8(c)). In the non-monsoon season, similar results for all the tests are found for 126 127.4 km^2 (67.8%), whereas similar results from the KRC and MK tests are

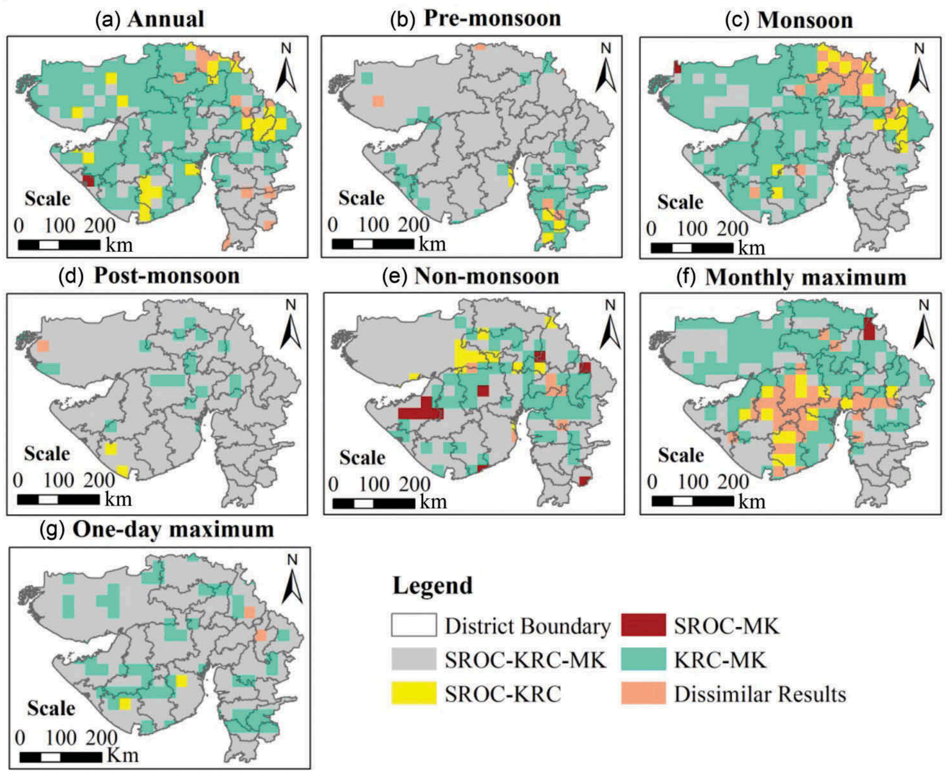

Figure 8. Distribution of similar and dissimilar results of SROC, KRC and MK tests in possible combinations over the study area.

obtained for 41 793.8 km^2 (22.5%) (Fig. 8(e)). The results of the KRC and MK tests for monthly maximum rainfall are similar for 76 295 km^2 (41%), with a predominance of similar results for all the tests over 73 563.5 km^2 (39.6%) (Fig. 8(f)). Finally, the results of the trend tests applied to one-day maximum rainfall depict similarity of all three test results in a major portion, i.e. 146 678 km^2 (78.9%), of the study area, whereas similar spatial trends are obtained from the KRC and MK test results for 36 370.2 km^2 (19.6%). Good harmony between the results of the KRC and MK tests has also been reported in earlier studies dealing with rainfall trend analysis (e.g. Machiwal and Jha 2008, 2016, Machiwal *et al.* 2016).

Based on the above discussion, it is clearly observed that, of five possible combinations of test results, two combinations, i.e. SROC-KRC-MK (similar results for all three tests) and KRC-MK (close agreement between the KRC and MK tests), are predominant in all the seven types of rainfall datasets. Also, it is seen that the trends identified in the pre-monsoon, post-monsoon and one-day maximum rainfalls by all three trend tests are consistent in a major portion of the study area, with the largest agreement over more than 92% of the area for the post-monsoon rainfall. Dissimilar results for all three trend tests are obtained for maximum (13.7%) in the case of monthly maximum and minimum (0.38%) for post-monsoon rainfall series.

From the earlier discussion of the trend results for seven types of rainfall series, it is inferred that significantly increasing trends are the most prominent in rainfall series of large magnitude, i.e. annual, monsoon and monthly maximum. These increasing rainfall trends are predominantly seen in the northeast, southwest and northwest parts of the study area. In contrast, decreasing rainfall trends are shown in the low-magnitude rainfall series of pre-monsoon, post-monsoon and non-monsoon seasons. However, decreasing trends are significant only in the pre-monsoon season (mainly in the southeast of the study area) and the non-monsoon season (in the northeast and southwest). Furthermore, it is interesting to note that the southeast part of the study area shows different behaviour from the rest of the area in the case of annual, pre-monsoon, monsoon and monthly maximum rainfall. It can be seen that decreasing rainfall trends prevail in the southeast, where the rainfall magnitudes generally remain higher than over the rest of the area.

4 Conclusions

This study employed rainfall datasets collected from satellite sensors, the Tropical Rainfall Measurement Mission (TRMM), at a high resolution of 0.25° × 0.25° for spatial modelling of rainfall trends. The novelty of the study lies in the development of a standard methodology for identifying the trends in the raster datasets by exploring the capabilities of the geographical information system (GIS) in implementing procedures for the three trend detection tests: the Spearman rank order correlation (SROC), Kendall's rank correlation (KRC) and the Mann-Kendall (MK) tests. The developed methodology was demonstrated by applying it to TRMM rainfall raster datasets for Gujarat State, India.

The mean annual rainfall in the study area varied from less than 400 mm to more than 1500 mm, with a SD of less than 200 mm to more than 350 mm. The coefficient of variation indicated large variability in annual rainfall in the areas experiencing relatively scant rainfall. The results of the KRC and MK tests showed a very high degree of coherence in spatial distribution of the identified trends for all seven types of rainfall datasets. Significantly increasing trends in the annual (39.9% area), monsoon season (38.4% area) and monthly maximum (23.5% area) rainfall were found to be prevalent in the Saurashtra-Kachchh region situated in the northwest and southwest parts of the study area, and in a fringe located in the northeast. In contrast, negative trends of annual rainfall were mainly confined within the southeast part of the study area, but none were statistically significant. In the pre-monsoon season, significantly decreasing rainfall trends were indicated in the southeast portion, whereas both increasing and decreasing rainfall trends were not statistically significant in the post-monsoon season except in some negligible scattered patches. In the non-monsoon season, significantly decreasing rainfall trends were observed in the Saurashtra region and the middle portion of the area (23% area). For one-day maximum rainfall, increasing trends were quite significant (7.9% area) in the middle of the Saurashtra region, a few patches in the northeast, and part of the Kachchh region; a major portion of the area (65.7%) showed non-significant increasing trends.

Comparative evaluation of the three trend tests revealed that their results were in fair agreement with each other in a major portion of the area for pre-monsoon, post-monsoon, non-monsoon and one-day maximum rainfall. However, the spatial distribution of the trends identified by the KRC and MK tests was found to be similar in a large portion of the study area for annual (56.3%), monsoon (49.4%) and monthly maximum (41%) rainfall. This suggests robustness of the KRC and MK tests in detecting trends in hydrological time series. Moreover, the methodology developed in this study could easily be employed in other parts of the world to examine rainfall trends and to model their spatial variability by utilizing satellite-based raster datasets at very high resolution.

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